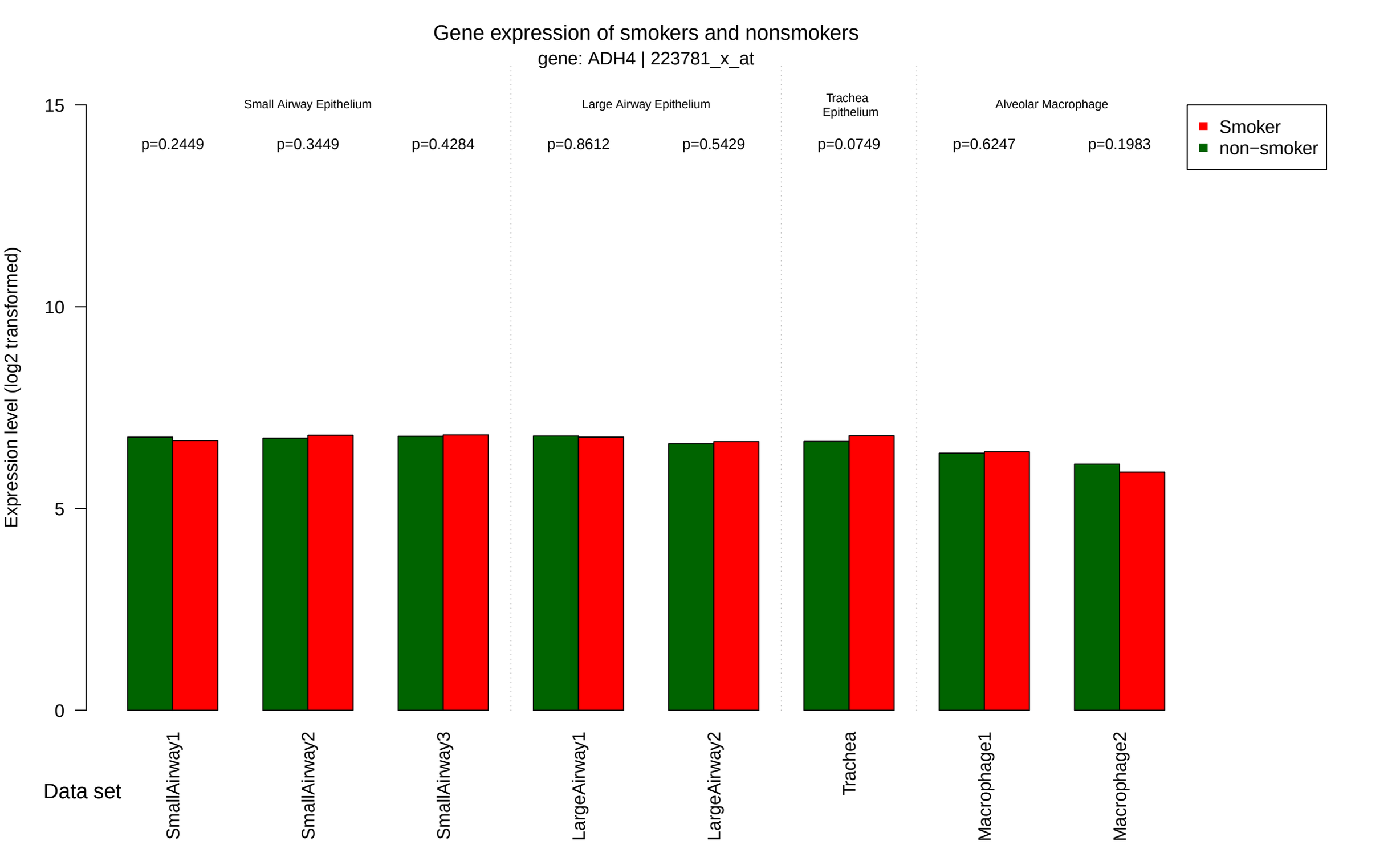
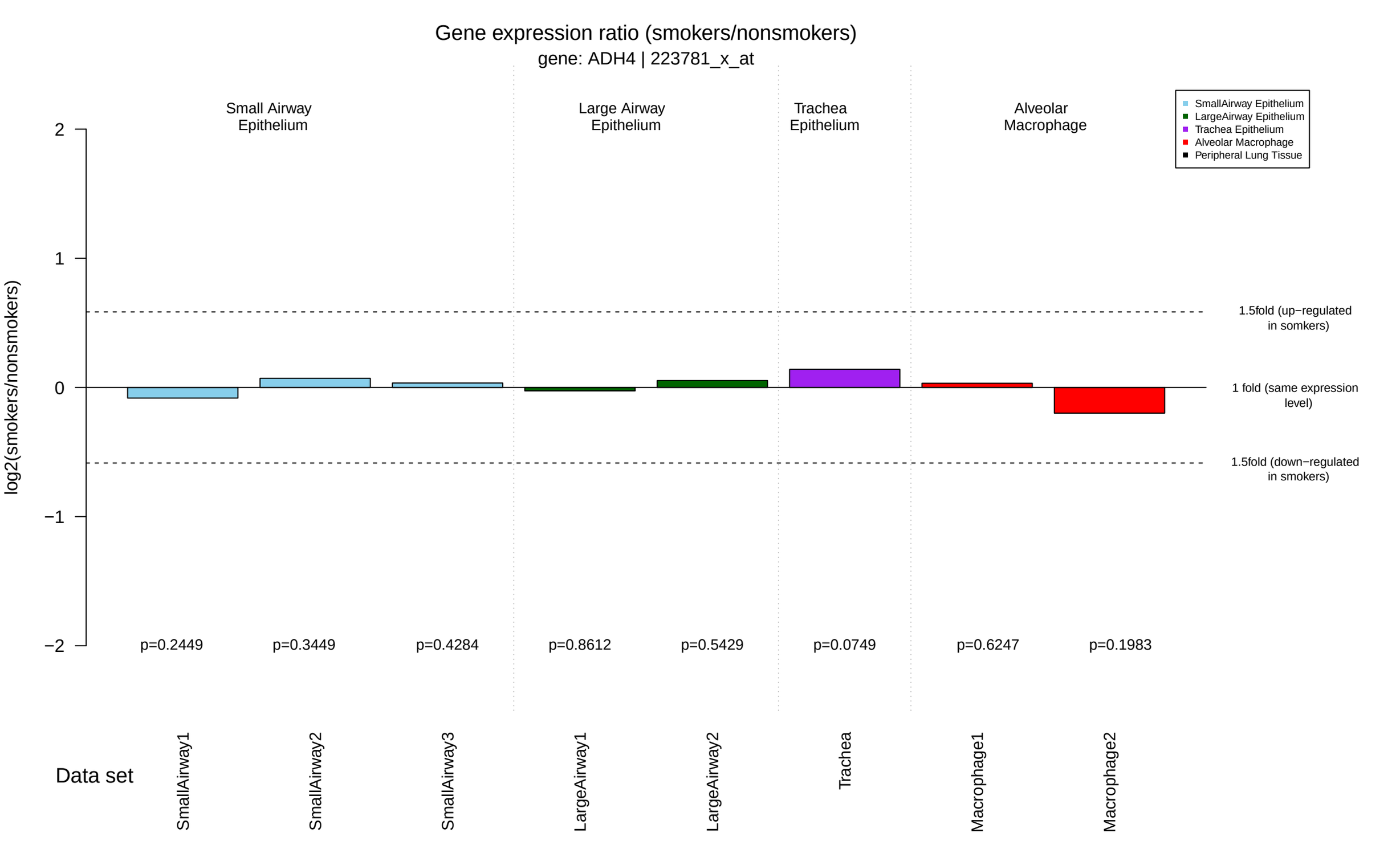
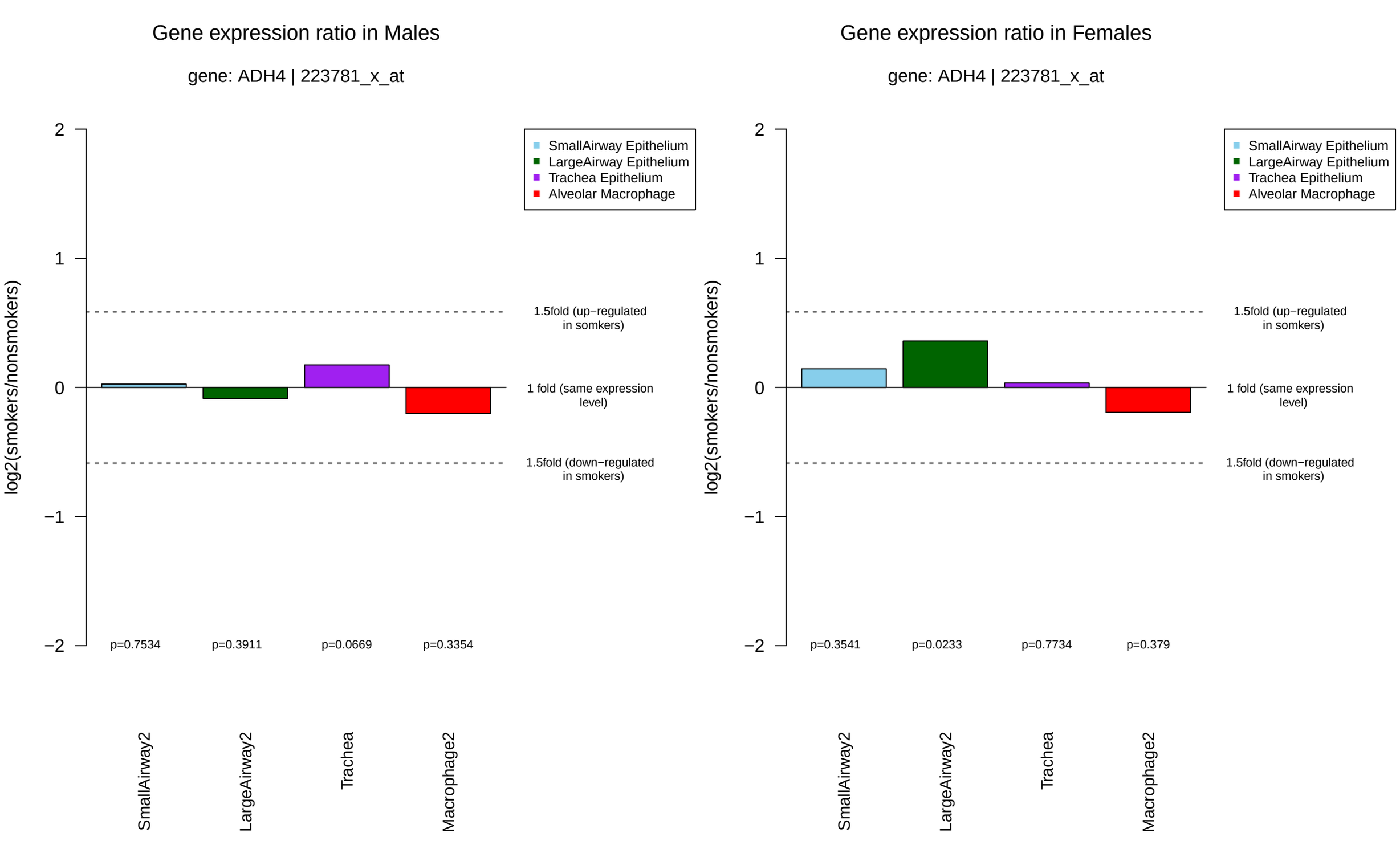
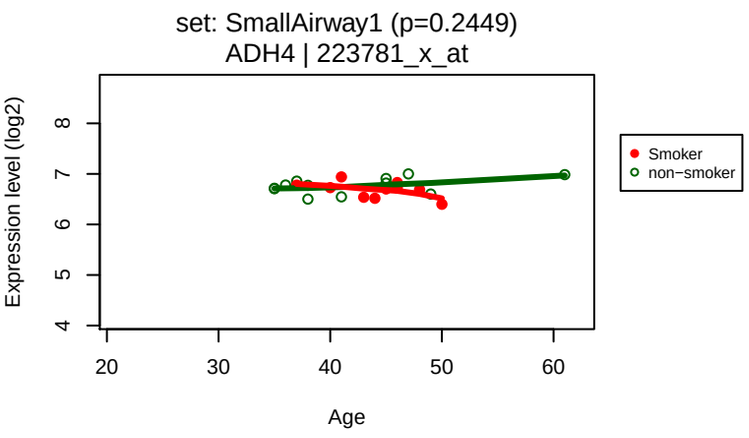
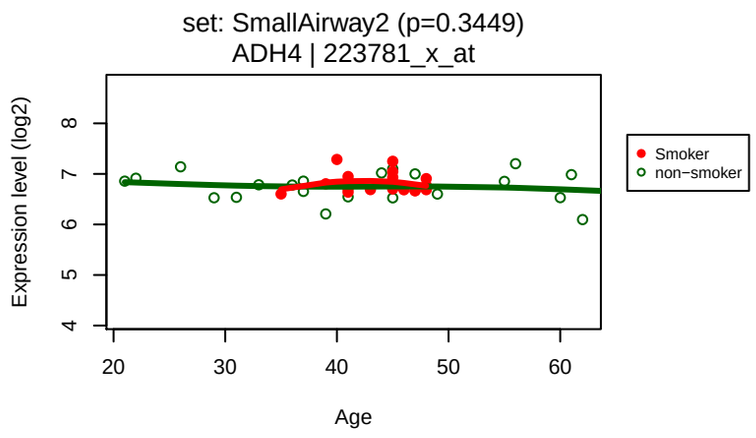
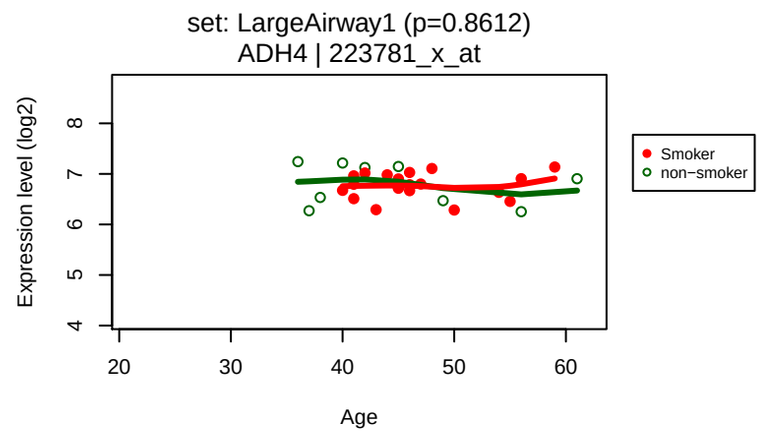

Gene expression of smokers and nonsmokers
gene: ADH4 | 223781_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2449
p=0.3449
p=0.4284
p=0.8612
p=0.5429
p=0.0749
p=0.6247
p=0.1983
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

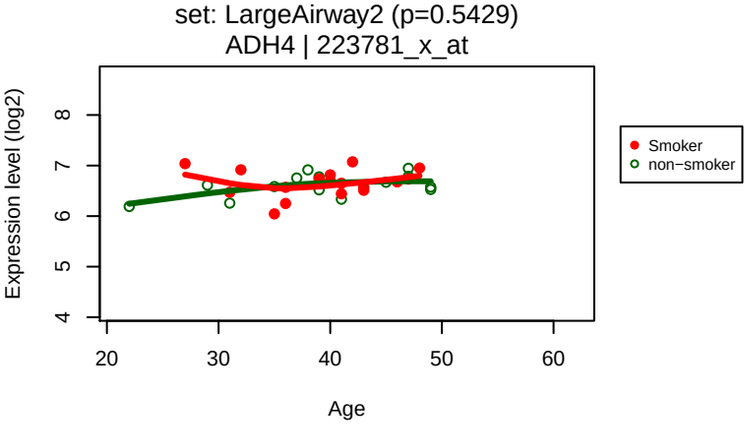

Gene expression ratio (smokers/nonsmokers)
gene: ADH4 | 223781_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2449
p=0.3449
p=0.4284
p=0.8612
p=0.5429
p=0.0749
p=0.6247
p=0.1983
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

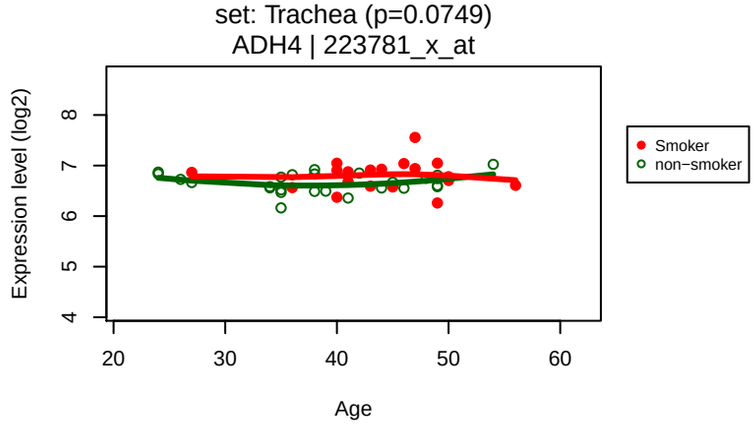

Gene expression ratio in Males
gene: ADH4 | 223781_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7534
p=0.3911
p=0.0669
p=0.3354
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADH4 | 223781_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3541
p=0.0233
p=0.7734
p=0.379
SmallAirway2
LargeAirway2
Trachea
Macrophage2

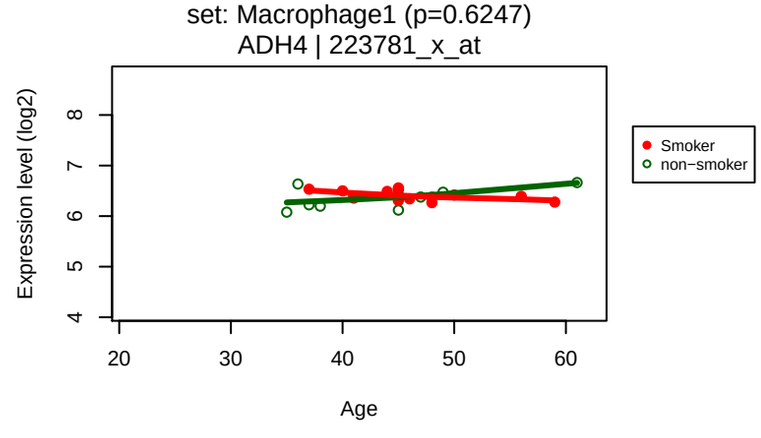

set: SmallAirway1 (p=0.2449)
ADH4 | 223781_x_at
Expression level (log2)
Age
Smoker
non−smoker

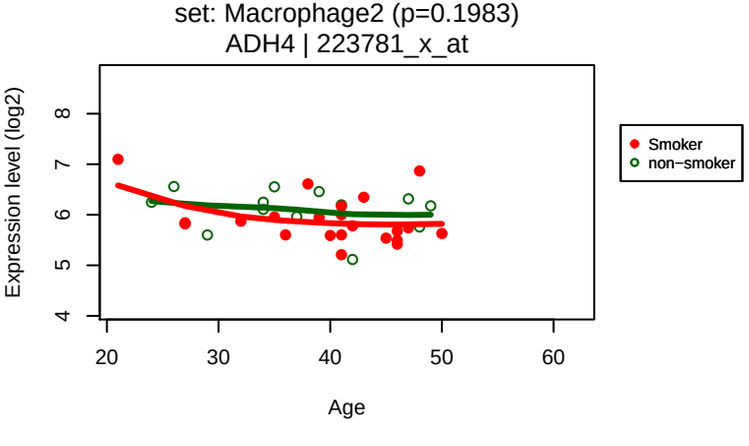

set: SmallAirway2 (p=0.3449)
ADH4 | 223781_x_at
Expression level (log2)
Age
Smoker
non−smoker

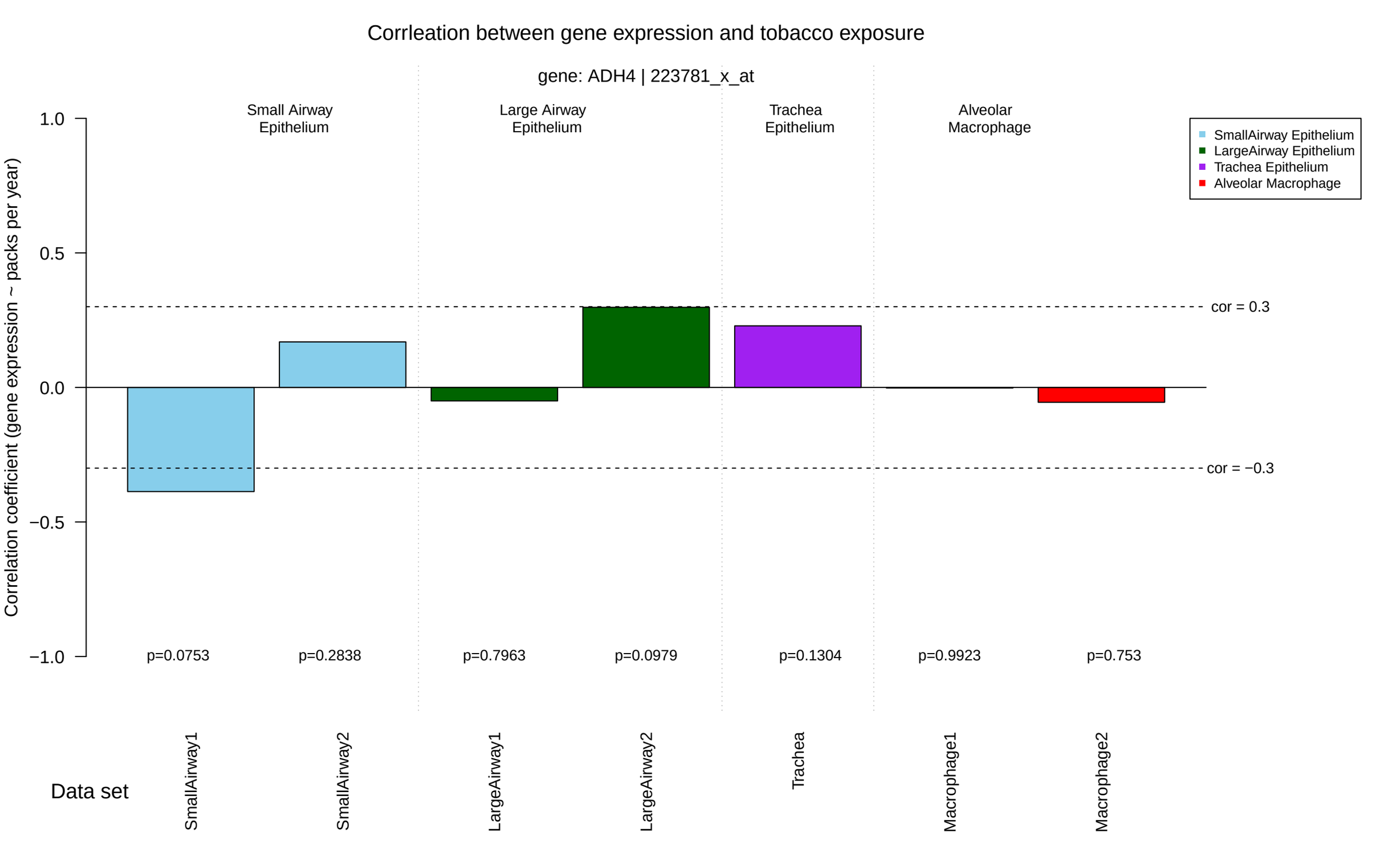

set: LargeAirway1 (p=0.8612)
ADH4 | 223781_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5429)
ADH4 | 223781_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0749)
ADH4 | 223781_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6247)
ADH4 | 223781_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1983)
ADH4 | 223781_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADH4 | 223781_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0753
p=0.2838
p=0.7963
p=0.0979
p=0.1304
p=0.9923
p=0.753
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2